TRIGONOMETRY

(त्रिकोणमिति)

SHEET-02

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

TRIGONOMETRY (त्रिकोणमिति)

CLASSROOM SHEET-02



If $\alpha + \beta = 90° \Rightarrow \alpha$ and β are complimentary.

$\tan\alpha \times \tan\beta = 1$, $\cot\alpha \times \cot\beta = 1$,

$\sin\alpha \times \sec\beta = 1$, $\cos\alpha \times \operatorname{cosec}\beta = 1$

1. $2(\sin 1° \times \sec 89°) + 3(\cos 11° \times \operatorname{cosec} 79°) + 5(\tan 21° \times \tan 69°) = ?$

SSC CPO 04/10/2023 (Shift-02)

(a) 20 (b) 12
(c) 11 (d) 10

2. The value of $\tan5° \tan25° \tan45° \tan65° \tan85°$ is equal to _____.

$\tan5° \tan25° \tan45° \tan65° \tan85°$ का मान _____ के बराबर है।

SSC CGL 24/07/2023 (Shift-2)

(a) 4 (b) 3
(c) 1 (d) 2

3. The value of $\cot 13° \cot 27° \cot 60° \cot 63° \cot 77°$ is:

$\cot 13° \cot 27° \cot 60° \cot 63° \cot 77°$ का मान है:

SSC CHSL 02/08/2023 (Shift-02)

(a) $\frac{1}{\sqrt{3}}$ (b) 0
(c) $\sqrt{3}$ (d) 1

4. Evaluate:/सरल करें।

$$7\left(\frac{\operatorname{cosec}24^\circ}{\sec66^\circ}\right)^3 + 8\left(\frac{\cot37^\circ}{\tan53^\circ}\right)^4 - \left(2\frac{\sec14^\circ}{\operatorname{cosec}76^\circ}\right)^2 + \left(-3\frac{\tan82^\circ}{\cot8^\circ}\right)^3$$

SSC Phase-XII 20/06/2024 (Shift-02)

(a) 40 (b) 2
(c) –16 (d) 10

5. Find the value of $\frac{\sec35^\circ}{\operatorname{cosec}55^\circ} - \frac{\cos25^\circ}{\sin65^\circ}$

$\frac{\sec35^\circ}{\operatorname{cosec}55^\circ} - \frac{\cos25^\circ}{\sin65^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 11/09/2024 (Shift-01)

(a) 0 (b) 1
(c) $\frac{1}{2}$ (d) 2

6. What is the value of $\frac{\cos50^\circ}{\sin40^\circ} + \frac{3\operatorname{cosec}80^\circ}{\sec10^\circ} - 2\cos50^\circ . \operatorname{cosec}40^\circ$?

$\frac{\cos50^\circ}{\sin40^\circ} + \frac{3\operatorname{cosec}80^\circ}{\sec10^\circ} - 2\cos50^\circ . \operatorname{cosec}40^\circ$ का मान ज्ञात कीजिए।

SSC CGL MAINS (08/08/2022)

(a) 3 (b) 4
(c) 2 (d) 5

7. The value of $\frac{\tan13^\circ\tan36^\circ\tan45^\circ\tan54^\circ\tan77^\circ}{2\sec^2 60^\circ(\sin^2 60^\circ - 3\cos60^\circ + 2)}$ is:

$\frac{\tan13^\circ\tan36^\circ\tan45^\circ\tan54^\circ\tan77^\circ}{2\sec^2 60^\circ(\sin^2 60^\circ - 3\cos60^\circ + 2)}$ का मान ज्ञात करें।

SSC CGL 20/08/2021 (Shift- 03)

(a) $-\frac{1}{4}$ (b) $-\frac{1}{10}$
(c) $\frac{1}{10}$ (d) $\frac{1}{4}$

8. $\left[\frac{\operatorname{cosec}31^\circ - \sec59^\circ}{(\sin18^\circ \div \cos72^\circ)} \times \frac{\tan26^\circ}{\cot64^\circ}\right] + \left[\frac{\tan30^\circ + \tan15^\circ}{(1 - \tan30^\circ\tan15^\circ)}\right]$ = _____.

SSC CHSL 05/07/2024 (Shift-04)

(a) 0 (b) –1
(c) 1 (d) 2

9. If $\sin(5x - 40^\circ) = \cos(5y + 40^\circ)$, then the value of $x + y$ is:

यदि $\sin(5x - 40^\circ) = \cos(5y + 40^\circ)$ तो $x + y$ का मान है:

SSC Phase-XII 24/06/2024 (Shift-03)

(a) 18° (b) 40°
(c) 15° (d) 20°

10. If $\sin(x + y) = \cos\{3(x + y)\}$, then, the value of $\cot\{2(x + y)\}$ is _____.

यदि $\sin(x + y) = \cos\{3(x + y)\}$ तो $\cot\{2(x + y)\}$ का मान है।

SSC CPO 29/06/2024 (Shift-02)

(a) –1 (b) 1
(c) –2 (d) 2

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11. If $\tan 2\theta = \cot(\theta - 36)^\circ$, where 2θ is an acute angle, then the value of θ is:
यदि $\tan 2\theta = \cot(\theta - 36)^\circ$, जहाँ 2θ एक न्यून कोण है, तो θ का मान ज्ञात कीजिए।

SSC CHSL 11/08/2023 (Shift-2)

(a) 18° (b) 36°
(c) 30° (d) 42°

12. If $\cos(2\theta + 54)^\circ = \sin\theta$, $0^\circ < (2\theta + 54)^\circ < 90^\circ$, then what is the value of $\frac{1}{\cot 5\theta + \sec\frac{5\theta}{2}}$?
यदि $\cos(2\theta + 54)^\circ = \sin\theta$, $0^\circ < (2\theta + 54)^\circ < 90^\circ$ है तो $\frac{1}{\cot 5\theta + \sec\frac{5\theta}{2}}$ का मान ज्ञात कीजिए।

SSC CGL 20/08/2021 (Shift- 02)

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{3}$
(c) $\frac{\sqrt{3}}{3}$ (d) $2\sqrt{3}$

13. If $\sin(5x - 25)^\circ = \cos(5y + 25)^\circ$, where $5x - 25^\circ$ and $5y + 25^\circ$ are acute angles, then the value of $(x + y)$ is:
यदि $\sin(5x - 25)^\circ = \cos(5y + 25)^\circ$ है, जहाँ $5x - 25^\circ$ और $5y + 25^\circ$ न्यूनकोण है, तो $(x + y)$ का मान क्या होगा।

SSC CGL 18/07/2023 (Shift-03)

(a) 50° (b) 40°
(c) 18° (d) 16°

14. If $\tan(5\theta - 10)^\circ = \cot(5\varphi + 20)^\circ$, then the value of $\theta + \varphi$ is :
यदि $\tan(5\theta - 10)^\circ = \cot(5\varphi + 20)^\circ$ है, तो $\theta + \varphi$ का मान क्या है?

SSC CHSL 10/08/2023 (Shift-2)

(a) 16° (b) 18°
(c) 15° (d) 20°

15. If $\sec(5\alpha - 15)^\circ = \text{cosec}(15 - 2\alpha)^\circ$, then the value of $\cos\alpha + \sin 2\alpha + \tan(1.5\alpha)$ is:
यदि $\sec(5\alpha - 15)^\circ = \text{cosec}(15 - 2\alpha)^\circ$ है तो $\cos\alpha + \sin 2\alpha + \tan(1.5\alpha)$ का मान ज्ञात कीजिए।

SSC CGL 20/08/2021 (Shift- 01)

(a) $\sqrt{2} + 1$ (b) $\sqrt{2} - 1$
(c) $\sqrt{3} - 1$ (d) $\sqrt{3} + 1$

16. If $\sin(20 + x)^\circ = \cos 60^\circ$, $0^\circ \le (20 + x)^\circ \le 90^\circ$, then find the value of $2\sin^2(3x + 15)^\circ - \text{cosec}^2(2x + 10)^\circ$.
यदि $\sin(20 + x)^\circ = \cos 60^\circ$, $0^\circ \le (20 + x)^\circ \le 90^\circ$ है तो $2\sin^2(3x + 15)^\circ - \text{cosec}^2(2x + 10)^\circ$ का मान ज्ञात कीजिए।

SSC CGL 20/08/2021 (Shift- 01)

(a) 3 (b) -3
(c) -2 (d) $-\frac{1}{3}$

17. If $\sin 3A = \cos(A + 10)^\circ$, where $3A$ is an acute angle then what is the value of $2\,\text{cosec}\frac{3A}{2} + 6\tan^2 3A - \frac{3}{2}\tan^2 3A$?
यदि $\sin 3A = \cos(A + 10)^\circ$ जहां $3A$ न्यूनकोण है तो $2\,\text{cosec}\frac{3A}{2} + 6\tan^2 3A - \frac{3}{2}\tan^2 3A$ का मान ज्ञात कीजिए।

SSC CGL 2019 Tier-II (15/10/2020)

(a) $\frac{7}{4}$ (b) $\frac{35}{2}$
(c) $\frac{17}{2}$ (d) 5

18. If $\cos\theta = \sin(2\theta) \neq 0$, what is the value of $\cos^4\theta + \sin^4\theta + \cos^3\theta + \sin^3\theta + \sin^2\theta + \cos^2\theta + \sin\theta + \cos\theta$?
यदि $\cos\theta = \sin(2\theta) \neq 0$, तो $\cos^4\theta + \sin^4\theta + \cos^3\theta + \sin^3\theta + \sin^2\theta + \cos^2\theta + \sin\theta + \cos\theta$ का मान क्या?

SSC Phase X 05/08/2022 (Shift- 03)

(a) $\frac{18 + 8\sqrt{3}}{7}$ (b) $\frac{8 + 7\sqrt{3}}{18}$
(c) $\frac{7 + 18\sqrt{3}}{8}$ (d) $\frac{18 + 7\sqrt{3}}{8}$

If $\alpha + \beta = 90^\circ$, then
$\sin^2\alpha + \sin^2\beta = 1$ &
$\cos^2\alpha + \cos^2\beta = 1$

19. Simplify the given expression.
दिए गए व्यंजक को सरल कीजिए।

$\frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos 53^\circ \sin 37^\circ$

SSC CHSL 07/08/2023 Shift-04

(a) 2 (b) -1
(c) -2 (d) 1

20. The value of $\sin^2 15^\circ + \sin^2 25^\circ + \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$ is:
$\sin^2 15^\circ + \sin^2 25^\circ + \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$ का मान है।

SSC CPO 29/06/2024 (Shift-03)

(a) 4 (b) 7
(c) $\frac{7}{2}$ (d) $\frac{7}{3}$

21. The value of $\frac{5\cos^2 62^\circ + 5\cos^2 28^\circ - 21}{7\sin^2 35^\circ + 7\sin^2 55^\circ + 1}$ is:
$\frac{5\cos^2 62^\circ + 5\cos^2 28^\circ - 21}{7\sin^2 35^\circ + 7\sin^2 55^\circ + 1}$ का मान ज्ञात कीजिए।

SSC CGL 20/04/2022 (Shift- 03)

(a) 3 (b) -2
(c) 2 (d) -3

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22. The value of $\frac{\sqrt{2}\tan(60-\theta)^\circ\tan(30+\theta)^\circ}{\sin^2(45+\theta)^\circ+\sin^2(45-\theta)^\circ}$:

$\frac{\sqrt{2}\tan(60-\theta)^\circ\tan(30+\theta)^\circ}{\sin^2(45+\theta)^\circ+\sin^2(45-\theta)^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 17/08/2021 (Shift- 02)

(a) $\frac{1}{\sqrt{2}}$ (b) 1

(c) 2 (d) $\sqrt{2}$

23. The value of $(\sin37^\circ\cos53^\circ+\cos37^\circ\sin53^\circ)-\frac{4\cos^237^\circ-7+4\cos^253^\circ}{\tan^247^\circ+4-\text{cosec}^243^\circ}$ is:

$(\sin37^\circ\cos53^\circ+\cos37^\circ\sin53^\circ)-\frac{4\cos^237^\circ-7+4\cos^253^\circ}{\tan^247^\circ+4-\text{cosec}^243^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 18/08/2021 (Shift- 01)

(a) 1 (b) –2

(c) 0 (d) 2

24. The value of

$\frac{4\tan^230^\circ+\sin^230^\circ\cos^245^\circ+sec^248^\circ-\cot^242^\circ}{\cos37^\circ\sin53^\circ+\sin37^\circ\cos53^\circ+\tan18^\circ\tan72^\circ}$

$\frac{4\tan^230^\circ+\sin^230^\circ\cos^245^\circ+sec^248^\circ-\cot^242^\circ}{\cos37^\circ\sin53^\circ+\sin37^\circ\cos53^\circ+\tan18^\circ\tan72^\circ}$ का मान ज्ञात कीजिए।

SSC CGL MAINS 29/01/2022

(a) $\frac{35}{48}$ (b) $\frac{59}{48}$

(c) $\frac{49}{24}$ (d) $\frac{35}{24}$

25. The value of $\frac{5\cos^262^\circ+5\cos^228^\circ-21}{7\sin^235^\circ+7\sin^255^\circ+1}$ is:

$\frac{5\cos^262^\circ+5\cos^228^\circ-21}{7\sin^235^\circ+7\sin^255^\circ+1}$ का मान ज्ञात कीजिए।

SSC CGL 20/04/2022 (Shift- 03)

(a) 3 (b) – 2

(c) 2 (d) – 3

26. The value of

$\frac{(\cos9^\circ+\sin81^\circ)(\sec9^\circ+\text{cosec}\,81^\circ)}{\text{cosec}^271^\circ+\cos^215^\circ-\tan^219^\circ+\cos^275^\circ}$ is:

$\frac{(\cos9^\circ+\sin81^\circ)(\sec9^\circ+\text{cosec}\,81^\circ)}{\text{cosec}^271^\circ+\cos^215^\circ-\tan^219^\circ+\cos^275^\circ}$ का मान ज्ञात कीजिए।

SSC CGL MAINS 03/02/2022

(a) 1 (b) 4

(c) – 3 (d) 2

27. If $x = \sec57^\circ$, then $\cot^233^\circ+\sin^257^\circ+\sin^233^\circ+\text{cosec}^257^\circ\cos^233^\circ+\sec^233^\circ.\sin^257^\circ$ is equal to :

यदि $x = \sec57^\circ$ है तो $\cot^233^\circ+\sin^257^\circ+\sin^233^\circ+\text{cosec}^257^\circ\cos^233^\circ+\sec^233^\circ.\sin^257^\circ$ का मान ज्ञात कीजिए।

SSC CGL TIER-II (16/10/2020)

(a) x^2+2 (b) $2x^2+1$

(c) x^2+1 (d) $\frac{1}{x^2+1}$

28. Find the value of the following.

निम्नलिखित का मान ज्ञात कीजिए।

$\frac{\sin67^\circ\cos37^\circ-\sin37^\circ\cos67^\circ}{\cos13^\circ\cos17^\circ-\sin13^\circ\sin17^\circ}$

SSC CHSL 27/05/2022 (Shift-03)

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{4}{\sqrt{3}}$

(c) $\frac{2}{\sqrt{3}}$ (d) 7

29. The value of $\frac{\sin^252^\circ+2+\sin^238^\circ}{4\cos^243^\circ-5+4\cos^247^\circ}$ is:

$\frac{\sin^252^\circ+2+\sin^238^\circ}{4\cos^243^\circ-5+4\cos^247^\circ}$ का मान ज्ञात करें।

SSC CPO 24/11/2020 (Shift-2)

(a) 3 (b) $\frac{1}{3}$

(c) $-\frac{1}{3}$ (d) –3

30. The value of

$\frac{2\sin^238^\circ\sec^252^\circ+\cos64^\circ\sin26^\circ+\sin^264^\circ}{\tan^223^\circ+\cot^223^\circ-\sec^267^\circ-\text{cosec}^267^\circ}$ is :

$\frac{2\sin^238^\circ\sec^252^\circ+\cos64^\circ\sin26^\circ+\sin^264^\circ}{\tan^223^\circ+\cot^223^\circ-\sec^267^\circ-\text{cosec}^267^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 2019 TIER-II (15/10/2020)

(a) – 2 (b) $\frac{3}{2}$

(c) 2 (d) $-\frac{3}{2}$

31. The value of

$\frac{3(\text{cosec}^226^\circ-\tan^264^\circ)+(\cot^242^\circ-\sec^248^\circ)}{\cot(22-\varnothing)^\circ-\text{cosec}^2(62+\varnothing)^\circ-\tan(\varnothing+68)^\circ+\tan^2(28-\varnothing)^\circ}$

$\frac{3(\text{cosec}^226^\circ-\tan^264^\circ)+(\cot^242^\circ-\sec^248^\circ)}{\cot(22-\varnothing)^\circ-\text{cosec}^2(62+\varnothing)^\circ-\tan(\varnothing+68)^\circ+\tan^2(28-\varnothing)^\circ}$

का मान ज्ञात कीजिए।

SSC CGL MAINS 03/02/2022

(a) 3 (b) 4

(c) –1 (d) –2

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32. The value of

$$\frac{3(\cot^2 47° - \sec^2 43°) - 2(\tan^2 23° - \cos ec^2 67°)}{\text{cosec}^2(68 + \theta)° - \tan(\theta + 61)° - \tan^2(22 - \theta)° + \cot(29 - \theta)°}$$

$$\frac{3(\cot^2 47° - \sec^2 43°) - 2(\tan^2 23° - \cos ec^2 67°)}{\text{cosec}^2(68 + \theta)° - \tan(\theta + 61)° - \tan^2(22 - \theta)° + \cot(29 - \theta)°}$$

का मान ज्ञात कीजिए।

SSC CGL MAINS 29/01/2022

(a) 1 (b) 0

(c) 5 (d) –1

33. The value of/का मान बताइए।

$$\frac{\sin 23°\cos 67° + \sec 52°\sin 38° + \cos 23°\sin 67° + \text{cosec}\, 52°\cos 38°}{\text{cosec}^2 20° - \tan^2 70°}$$

SSC CGL 11/04/2022 (Shift- 01)

(a) 3 (b) 4

(c) 2 (d) 0

ANSWER KEY

1.(d)	**2.**(c)	**3.**(a)	**4.**(c)	**5.**(a)	**6.**(c)	**7.**(c)	**8.**(c)	**9.**(a)	**10.**(b)
11.(d)	**12.**(c)	**13.**(c)	**14.**(a)	**15.**(d)	**16.**(b)	**17.**(b)	**18.**(d)	**19.**(a)	**20.**(c)
21.(b)	**22.**(d)	**23.**(d)	**24.**(b)	**25.**(b)	**26.**(d)	**27.**(a)	**28.**(a)	**29.**(d)	**30.**(d)
31.(d)	**32.**(d)	**33.**(a)							

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CALCULATION TECHNIQUES

TECHNIQUE - 01

♦ Cube Root of Non-perfect cube ♦

$$\sqrt[3]{N} = A \pm \frac{B}{3.A^2}$$

First find a number which is a perfect cube near to the number N, let it be A and the difference between N and A^3 is B.

Note: '+' sign is used if $N > A^3$ & '–' sign is used if $N < A^3$

Ex. $\sqrt[3]{60} = 4 - \frac{4}{3 \times 4^2} = \frac{188}{48} = 3.92$

Ex. $\sqrt[3]{72} = 4 + \frac{8}{3 \times 4^2} = \frac{200}{48} = 4.17$

♦ Do it Yourself ♦

(i) $\sqrt[3]{10}$ = ...

(ii) $\sqrt[3]{66}$ = ...

(iii) $\sqrt[3]{118}$ = ...

(iv) $\sqrt[3]{226}$ = ...

(v) $\sqrt[3]{525}$ = ...

(vi) $\sqrt[3]{1365}$ = ...

(vii) $\sqrt[3]{1500}$ = ...

TECHNIQUE - 02

♦ Square of two digit number ♦

$(ab)^2 = a^2/2ab/b^2$ (+ Carry, + Carry; 1 digit, 1 digit)

- **Square of 14**

Here, a = 1 & b = 4

$14^2 = 1^2/2.1.4/4^2 = 1/8/16 = 196$

- **Square of 23**

Here, a = 2 & b = 3

$23^2 = 2^2/2.2.3/3^2 = 4/12/9 = 529$

♦ Do it Yourself ♦

(i) 18^2 = ... (ii) 23^2 = ...

(iii) 27^2 = ... (iv) 34^2 = ...

(v) 38^2 = ... (vi) 43^2 = ...

(vii) 57^2 = ... (viii) 64^2 = ...

(ix) 66^2 = ... (x) 75^2 = ...

(xi) 78^2 = ... (xii) 84^2 = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
TRIGONOMETRY SHEET - 02						
If $\beta + \alpha = 90° \alpha \Rightarrow$ and β are complimentary. $\tan\alpha \times \tan\beta = 1$, $\cot\alpha \times \cot\beta = 1$, $\sin\alpha \times \sec\beta = 1$, $\cos\alpha \times \operatorname{cosec}\beta = 1$						
If $\beta + \alpha = 90°$, then $\sin2\alpha + \sin2\beta = 1$ & $\cos2\alpha + \cos2\beta = 1$						